

Design Verification Plan & Report (DVP&R;) — Virtual Test

Case ID: t3_r1_mimo
Document Number: VBF-T3R1MIMO-DVPR-001
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1. Test Plan

Test ID	Test Description	Criteria	Method
T01	1C constant-current discharge capacity	$\geq 2.0\text{ Ah}$	PyBaMM DFN simulation, Chen2020
T02	5C discharge capacity retention	$5C_{\text{cap}} / 1C_{\text{cap}} \geq 0.95$	PyBaMM DFN simulation
T03	4C fast charge — lithium plating check	Anode potential $\geq 0\text{V}$ at all times	PyBaMM SPMe + plating module, 45°C
T04	4C fast charge — maximum temperature	$T_{\text{max}} \leq 333.15\text{ K (60°C)}$	PyBaMM lumped thermal model
T05	Power density	$\geq 4000\text{ W/kg}$	Derived from $V_{\text{OC}}^2 / (4 \cdot \text{DCR}) / \text{mass}$
T06	Energy density (informational)	—	Contract-caliber integral

2. Test Results

Test ID	Target	Result	Status	Source
T01	$\geq 2.0\text{ Ah}$	2.31 Ah	✓ PASS	r4_enhanced_1c.json
T02	≥ 0.95	0.968	✓ PASS	r4_enhanced_1c.json + r4_enhanced_5c.json
T03	No plating	Anode min +0.005V	✓ PASS	r4_enhanced_4c.json
T04	$\leq 333.15\text{ K}$	327.6 K	✓ PASS	r4_enhanced_4c.json
T05	$\geq 4000\text{ W/kg}$	65,280 W/kg	✓ PASS	r4_energy.json
T06	Informational	308.4 Wh/kg	—	r4_energy.json

3. Test Conditions

- **Electrolyte**: Enhanced transport ($\sigma=1.5\text{ S/m}$, $D=7.5\times10^{-14}\text{ m}^2/\text{s}$, $t_{\text{eff}}=0.45$)
- **Thermal management**: $h=50\text{ W/m}^2\cdot\text{K}$ (aggressive cooling)
- **Simulation model**: DFN for discharge (1C, 5C); SPMe for 4C charge (DFN fallback)
- **Ambient temperature**: 298.15 K (1C/5C), 318.15 K (4C charge at 45°C)

4. Notes

- Virtual testing only — no physical cells fabricated or tested
- Electrolyte transport parameters are domain estimates, not experimentally verified
- True DFT/MD endorsement skipped (real_compute=false per task configuration)
- Aging/cycle-life testing not performed (not in task scope)